AWS REFERENCE ARCHITECTURE

EKS Dev/Test Platform

Single-page reference: edge → compute → app → data → observability → security → backup, with bootstrap & pipeline.

DEV/TEST us-east-1

AWS CLOUD · ACCOUNT: CLOUDOPS-DEV

EDGE
public ingress

Route 53
api.dev.cloudops.io
A/AAAA · TTL 60

HTTPS

WAFv2
managed rules + rate-limit
AWS-Common · Bot-Ctrl · 2k r/5m

ALB
LBC-managed (IngressClass=alb)
public · TLS 1.3

ACM

ACM Certificate
*.dev.cloudops.io

COMPUTE
VPC · 3 AZs

VPC 10.40.0.0/16 3 AZs · NAT-per-AZ · private nodes & RDS · public ALB only

AZ US-EAST-1A

PUBLIC SUBNET

ALB target
10.40.0.0/20
public

NAT

NAT GW 1a
EIP-attached
egress

PRIVATE SUBNET · NODES

Node 1
m6i.large · ON_DEMAND
k8s-node

PRIVATE SUBNET · DATA

RDS primary
db.t3.medium
multi-az · writer

AZ US-EAST-1B

PUBLIC SUBNET

ALB target
10.40.16.0/20
public

NAT

NAT GW 1b
EIP-attached
egress

PRIVATE SUBNET · NODES

Node 2
m6i.large · ON_DEMAND
k8s-node

PRIVATE SUBNET · DATA

RDS standby
db.t3.medium
multi-az · reader

AZ US-EAST-1C

PUBLIC SUBNET

ALB target
10.40.32.0/20
public

NAT

NAT GW 1c
EIP-attached
egress

PRIVATE SUBNET · NODES

Node 3
m6i.large · ON_DEMAND
k8s-node

PRIVATE SUBNET · DATA

RDS read repl.
db.t3.medium
async

EKS CLUSTER · CLOUDOPS-DEV control plane (managed) · k8s 1.30 · OIDC issuer enabled

NS: CLOUDOPS application

- OA — **order-api** — Deployment · 3 replicas — HPA · PDB · NetPol
- SA — **order-api-sa** — IRSA · iam-role/order-api — OIDC

NS: MONITORING kube-prometheus-stack

- P — **kube-prometheus** — scrape :9090 — STS
- AM — **Alertmanager** — routes → SNS/Slack

NS: GATEKEEPER-SYSTEM OPA policy

- GK — **Gatekeeper** — admission webhook — constraint

NS: EXTERNAL-SECRETS ESO

- ES — **External Secrets** — SecretStore: SecretsManager — CRD

ECR pull-through cache
ecr-public/* · gcr.io/* · quay.io/*
no NAT to public ECR

VPC ENDPOINT → nodes pull images via PrivateLink — no egress to public ECR

APP
workload

order-api
Deployment · cloudops ns
HPA · PDB · NetworkPolicy

:80→909

- H — **/health** — liveness probe
- R — **/ready** — readiness probe
- M — **/metrics:9090** — Prom scrape target

JDBC

orders DB
postgres · cloudops_orders
SSL required

DATA
persistence

RDS PostgreSQL
db.t3.medium · 100GB gp3
Multi-AZ · 15.7 · enc-at-rest

ENCRYPT

KMS CMK
alias/cloudops-rds
rotation: yearly

CREDS

Secrets Manager
cloudops/orders/db
rotation: 30d

Rotation Lambda
SecretsManagerRDSPostgreSQL
VPC · priv subnet

OBSERVABILITY
metrics · logs · traces

- **kube-prometheus** — scrapes :9090 — STS · PVC
- L — **Loki** — logs · S3 backend — multi-tenant
- T — **Tempo** — traces · OTLP — ingester
- M — **Mimir** — long-term metrics — S3 chunks
- **Grafana** — unified alerting · SSO — dashboards · alerts
- AM — **Alertmanager** — routes · dedup — SNS · Slack
- Y — **yace exporter** — CloudWatch → Prometheus — poll 60s → **CloudWatch** — ALB, RDS, EKS metrics — source

SECURITY
continuous

- **GuardDuty** — EKS audit + runtime — enabled
- **Security Hub** — CIS · AWS FSBP — agg
- **AWS Config** — conformance pack — drift
- **CloudTrail** — org trail · S3 + KMS — mgmt+data
- OPA — **Gatekeeper** — OPA constraints — k8s admission
- ESO — **External Secrets** — SecretStore CRDs — SM ← k8s

BACKUP
recovery

AWS Backup vault
cloudops-dev-vault · KMS-enc
cross-region copy

Backup plan
daily 02:00 · weekly Sun · monthly 1st
ret 7/30/365d

PROTECT

RDS snapshots
cloudops-orders

S3 backup target
cloudops-dev-backups
versioned · obj-lock

BOOTSTRAP

persistent · pre-cluster

BOOTSTRAP ACCOUNT-LEVEL (PERSISTENT ACROSS REBUILDS)

- **S3 state bucket** — tf state · KMS · versioned — `locked`
- **OIDC provider** — token.actions.githubusercontent.com — `thumbprint`
- **GitHub Actions IAM role** — role/github-deploy · sub: repo:* — `trust` `OIDC`
- **ECR pull-through cache** — ecr-public/* · upstream rules — `nodes ← here`

PIPELINE

GitHub Actions

GITHUB ACTIONS → STAGED TERRAFORM APPLIES

1. **bootstrap + networking + dns** — terraform apply · auto-approve on main
2. **eks** — terraform apply · auto-approve on main
3. **ingress** — terraform apply · auto-approve on main
4. **observability** — terraform apply · auto-approve on main
5. **security** — terraform apply · auto-approve on main
6. **database** — terraform apply · auto-approve on main
7. **self_healing · slo · finops** — terraform apply · auto-approve on main

LEGEND — sync request path · --- async / IAM trust / event · VPC / subnet / namespace · service tile · K non-AWS component (typographic)